Home WiFi Network Configuration using Cisco Packet Tracer

Objective

Create a home WiFi network in Cisco Packet Tracer consisting of one wireless router and three PCs. Configure static IP addresses, verify connectivity, and document the setup process.

Step 1: Create a New Packet Tracer Project

1. Open Cisco Packet Tracer.
2. Select **File** → **New**.
3. Save the project as:

Plain Text

HomeWiFiNetwork.pkt

Step 2: Add Network Devices

Add the following devices from the device panel:

- 1 Wireless Router
- 3 PCs

Rename the PCs as:

Plain Text

PC1

PC2

PC3

Step 3: Connect Devices

Connect all three PCs to the wireless router.

Network Topology

Plain Text

Wireless Router

192.168.10.1

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| | |

PC1 PC2 PC3

192.168.10.10 192.168.10.11 192.168.10.12

Step 4: Configure Router

Configure the Wireless Router with:

Parameter	Value
Router IP	192.168.10.1
Subnet Mask	255.255.255.0
Default Gateway	N/A

Step 5: Configure Static IP Addresses

PC1

Setting	Value
IP Address	192.168.10.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1

PC2

Setting	Value
IP Address	192.168.10.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1

PC3

Setting	Value

IP Address	192.168.10.12
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1

Step 6: Verify Connectivity

Open Command Prompt on each PC and run the following commands.

From PC1

BAT

ping 192.168.10.11

ping 192.168.10.12

ping 192.168.10.1

From PC2

BAT

ping 192.168.10.10

ping 192.168.10.12

ping 192.168.10.1

From PC3

BAT

ping 192.168.10.10

ping 192.168.10.11

ping 192.168.10.1

Expected Result

Plain Text

Reply from 192.168.10.x

Packets: Sent = 4, Received = 4, Lost = 0

This confirms successful communication between all devices and the router.

Step 7: Screenshots to Include

For submission, include the following screenshots:

1. Complete Network Topology
2. Router Configuration Window
3. PC1 IP Configuration
4. PC2 IP Configuration
5. PC3 IP Configuration
6. Successful Ping Results

Conclusion

A home WiFi network was successfully created using Cisco Packet Tracer. Three PCs were connected to a wireless router, assigned static IP addresses within the same subnet, and configured with the router as the default gateway. Ping tests confirmed successful connectivity between all devices and the router.